

REPORT OF PARTICIPATION IN THE ATRIUM TRAINING SCHOOL: COMPUTER VISION IN ARCHAEOLOGY (14-18 SEPTEMBER 2026)

Miguel Carrero-Pazos

University of Santiago de Compostela

Monday, 14 September 2026: Kick-off and Introduction to Computer Vision

The first day of the ATRIUM Training School focused on a kick-off which covered organizational things of the training school, the objectives and the activities which were going to take place. We had to present ourselves and we were introduced to the teachers of the Summer School and Administration staff.

The first session introduced the topic of computer vision in Archaeology and its main applications in archaeological Science. Petr Pajdla showed how computational methods can be used to process and analyse images which identify archaeological objects and detect visual patterns, and why this is relevant to archaeological research. We were introduced to several case studies which successfully applied computer vision in archaeological projects, such as prehistoric mound detection.



Fig. 1. First kick-off day with the presentation of Petr Pajdla on Computer Vision in Archaeology.

This introduction was useful for understanding the general principles behind image-based analysis and the different stages involved in a typical computer vision workflow. It also provided an opportunity to consider how these approaches might

be applied to archaeological questions and datasets. Another important part of the day was the afternoon, where Ronald Harasim helped us to understand how to code (and not code) with Agents.

Python fundamentals were covered by Filip Hájek in the afternoon, as a starting point for programming. As a researcher who primarily uses R for statistical analysis and spatial modelling, I found the introduction to Python particularly useful. Python is becoming widely used in Archaeology, computer vision and machine learning, and gaining experience with this environment will help me expand the computational tools available for my research.

The first day concluded with an informal ice-breaker activity, which provided an opportunity to meet the other participants and exchange information about our research interests.



Fig. 2. Ice-breaker party, where we had a great time.

Tuesday, 15 September 2026: Archaeological Artefact Datasets and Image Annotation

The second day focused on archaeological datasets and the preparation of images for computer vision applications. In the morning sessions, Petr addressed the characteristics of image datasets and the importance of organising and annotating them in CVAT. As we were able to see, annotation is one of the main steps in machine learning: identifying objects or features within images and recording this information in a consistent format so that it can be used to train or evaluate a model.

The practical session also introduced OpenCV, a widely used computer vision library, together with detection masks and image augmentation.

During the evening, we had the opportunity to see participant's presentations, where a selection presented their case studies.



Fig. 3. Some of the participants gave very nice presentations.



Fig. 4. Tuesday finished with a tour around the city of Brno. Lovely walk!

Wednesday, 16 September 2026: Microscopic images dataset. Theory, image data & processing; Practice: data standardization & augmentation

Wednesday provided an interesting introduction on small-scale visual patterns. The practical session focused on machine learning applied to microscopic use-wear analysis of archaeological stone tools. Using microscope images, we explored how different processing techniques can be used to standardise and improve archaeological datasets before applying machine-learning methods. We also experimented with image binarisation to distinguish polished from unpolished surfaces, denoising techniques to remove unwanted artefacts, and data augmentation to increase the variability of relatively small archaeological image datasets. We also analysed micromorphology images from sediment thin sections. We used CLAHE for loading and visualising the images and improving their local contrast, later using a watershed algorithm to identify and separate individual regions, such as grains, fragments, clays or other microscopic features. Finally, canny edge detection was used to identify boundaries and transitions within the images.

During the afternoon we had a tour visit to the Institute laboratories, with a great introduction by specialized staff of the research the personal is doing.



Fig. 5. Visit to the conservation facilities in the Institute. We had a nice introduction to some equipment such as this TopoScan microscope for analytical images of archaeological objects.

Thursday, 17 September 2026: Satellite Images dataset: theory and practice; Rock art dataset: theory and practice

The fourth day of the Summer School was the most relevant to my research because it focused on satellite imagery and rock art detection. The morning sessions introduced the use of satellite imagery in archaeological research and explored how computer vision methods can be applied to remotely sensed data. As remote

sensing is already an important part of my research, particularly using LiDAR, GIS, and spatial analysis to investigate archaeological landscapes, the training provided an opportunity to explore how image-based methods could complement the approaches I already use. We went through a Google colab notebook which introduced us a complete workflow for working with Sentinel-2 imagery, starting with the visualisation of multispectral bands and different image composites to identify archaeological features. We then explored vegetation indices, such as NDVI, and techniques designed to enhance subtle anomalies in the landscape. The practical session also introduced some image filtering and machine-learning classification, to distinguish potential archaeological cropmarks from the surrounding landscape. This provided a useful example of how satellite imagery, image processing, and machine learning can be combined to support the identification of archaeological features over relatively large areas.

The programme also included a session on rock art datasets in the evening. The session focused on the use of computer vision and machine learning for the detection of rock art, using a practical case study based on archaeological cupmarks. Again, we worked through a Google Colab notebook using a pre-trained YOLO segmentation model, which was labelled on an archaeological dataset. The workflow covered the preparation and organisation of training, validation, and test data, the training of the model, and the subsequent detection of cupmarks in new images, as we see in figure 6:

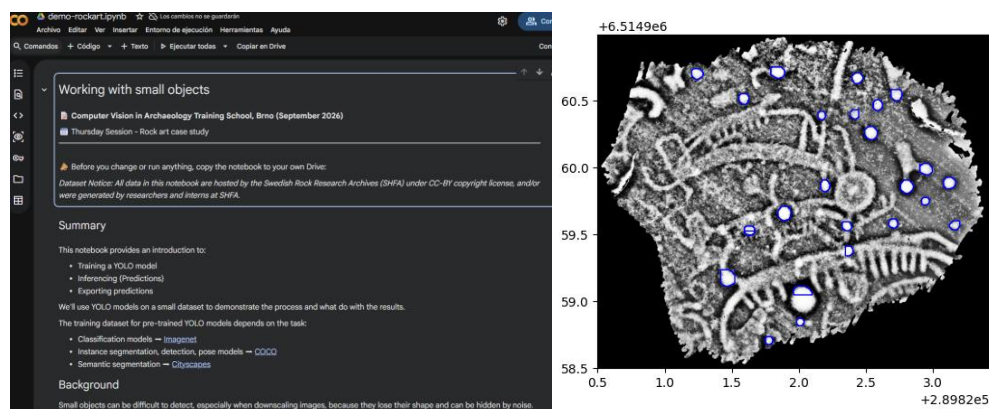


Fig. 6. The Google colab Notebook used for following the class and one of the results we got.

This day finished with a visit to the Institute's very nice Archive.



Fig. 7. Visit to the library/archive of the Institute, where we had the opportunity to see some of their old archaeological reports.

Friday, 18 September 2026: Coins dataset and closing of the summer school

The final day focused on how to apply machine learning detection algorithms in a coin dataset, with the problems and bias we might face. The training school concluded with a closing session and discussion, providing an opportunity to reflect on the topics covered during the week and their possible applications in future research.

Conclusions from the Summer School

Overall, the ATRIUM Training School was a valuable opportunity to expand my knowledge of computer vision and machine learning in archaeology. The combination of theoretical sessions with practical exercises into real-case archaeological datasets, and the discussions with teachers and participants, provided an interesting overview on computer vision in Archaeology. The Summer School covered both the technical aspects of working with real images and the methodological issues that need to be considered when applying machine learning to archaeological research. For my own research, the training was particularly useful because it connected computer vision with remote sensing, spatial analysis, and the study of archaeological landscapes. It also provided an opportunity to develop my Python skills and to explore methods that could complement the GIS, LiDAR, and R-based workflows I already use. The knowledge gained during this summer school will be useful for me in future applications of computer vision to archaeological datasets, especially those involving the detection and analysis of features in extensive image and remote sensing collections.



Fig. 7. Participants and teachers of the Summer School.